SINGAPORE JUNIOR PHYSICS OLYMPIAD 2021

GENERAL ROUND

1 July 2021 — 1700 – 1830

Time Allowed: **1.5 Hours**

INSTRUCTIONS

1. **Read the instructions on this page** but **DO NOT TURN the page** until you are told to do so by the invigilator.

2. This paper contains **50** multiple choice questions and **17** printed pages (including this cover page). Each question or incomplete statement is followed by five suggested answers or completions. **Select only the best** in each case and then **shade** the corresponding bubble on the answer sheet.

3. Use **2B pencil** only to shade the bubbles on the answer sheet, and make sure any stray markings are properly erased.

4. At the end of the test, please **submit** the **answer sheet** and the **question paper (attach all rough paper** used during the competition to the question paper). **Only** the **answer sheet** will be **marked**. Answers written anywhere else will not be marked.

5. **Fill in your index number in the space labelled as index number on the answer sheet now.** Write your **name** and **school** on the **answer sheet** and **question paper** now.

6. **Scientific calculators** are **allowed**. **Graphing** calculators are **not** allowed.

7. Answer **ONLY** questions you are confident of. Marks will be deducted for wrong answers. **A question left unanswered (blank) will score a higher mark than a wrong answer.**

8. A general data sheet is given in the following page. You may **detach the data sheet when the competition starts** so that you can refer to it easily.

Name: ______________________________

School: ______________________________

GENERAL DATA SHEET

Acceleration due to gravity at the surface of Earth, g = 9.80 ms $^{-2}$ = $|g|$

Universal gravitational constant; G = 6.67×10^{-11} kg $^{-1}$ m ^{3} s $^{-2}$

Universal gas constant, R = 8.31 J mol $^{-1}$ K $^{-1}$

Vacuum permittivity, ε_0 = 8.85×10^{-12} C ^{2} N $^{-1}$ m $^{-2}$

Vacuum permeability, μ_0 = $4\pi\times10^{-7}$ T m A $^{-1}$

Atomic mass unit, u = 1.66×10^{-27} kg

Speed of light in vacuum, c = 3.00×10^{8} m s $^{-1}$

Speed of sound in air, v_s = 340 ms $^{-1}$

Charge of electron, $e = 1.60\times10^{-19}$ C

Planck's constant, $h = 6.63\times10^{-34}$ J s

Mass of the Earth, $M_E = 5.97\times10^{24}$ kg

Mass of electron, $m_e = 9.11\times10^{-31}$ kg = 0.000549u

Mass of proton, $m_p = 1.67\times10^{-27}$ kg = 1.007u

Mass of deuteron, $m_d = 3.34\times10^{-34}$ kg = 2.014u

Rest mass of alpha particle, $m_\alpha = 4.003\,u$

Boltzmann constant, $k = 1.38\times10^{-23}$ J K $^{-1}$

Avogadro's number, $N_A = 6.02\times10^{23}$ mol $^{-1}$

Standard atmosphere pressure, $P_0 = 1.01\times10^{5}$ Pa

Density of water, $\rho_w = 1000$ kg m $^{-3}$

Specific heat (capacity) of water, $c_w = 4.19\times10^{3}$ J kg $^{-1}$ K $^{-1}$

Stefan-Boltzmann constant, $\sigma = 5.67\times10^{8}$ W m $^{-2}$ K $^{-4}$

Radius of the sun, $R_S = 6.96\times10^{5}$ km

Radius of the earth, $R_E = 6.37\times10^{3}$ km

Distance between sun and earth, $d_{SE} = 1.5\times10^{8}$ km

Acceleration due to gravity at the sun's surface, $g_S = 28.02$ g

Temperature on the surface of the sun, $T_S = 5780$K

1. Consider a sphere of mass 100 kg with a radius of 1.00 m falling through a strange medium which provides a dissipative force of $Akv^2 + bv$ where A is the cross-sectional area of the object. Given that k = 0.0400 kgm^{-3} and b = 0.400 kgs^{-1}, find its terminal velocity.

   A. 26.6 m/s
   B. 152 m/s
   C. 90.0 m/s
   D. 88.4 m/s
   E. 17.7 m/s
   F. 95.3 m/s
   G. 86.8 m/s
   H. 133 m/s

2. A 10 kg ball is launched up a rough inclined surface, 30° to the horizontal, with a frictional coefficient of 0.30. Along the way up the ramp, it compresses a spring of spring constant 1000 Nm^{-1} by 0.10 m. The distance travelled up the ramp is 5.0 m. What is the initial speed of the ball?

   A. 76 ms^{-1}
   B. 8.7 ms^{-1}
   C. 4.5 ms^{-1}
   D. 12 ms^{-1}
   E. 7.0 ms^{-1}
   F. 3.6 ms^{-1}
   G. 87 ms^{-1}
   H. 17 ms^{-1}

3. A 0.1 kg ball strikes a wall with a speed 4 ms^{-1} with an angle of 60° from the normal. It then reflects elastically. Assuming that the contact time is 0.01 s, what is the average force between the wall and the ball during the time of contact?

   A. 20 N
   B. 70 N
   C. 40 N
   D. 30 N
   E. 80 N
   F. 60 N
   G. 50 N
   H. 90 N

4. You discovered a new planet and found that the gravitational field of the planet is $3.0g$ and the density of the planet is 2 times that of Earth. Deduce its radius in terms of Earth's radius, R.

   A. 6.0 R
   B. 18 R
   C. 12 R
   D. 0.75 R
   E. 4.0 R
   F. 9.0 R
   G. 0.67 R
   H. 1.5 R

5. An object was on the x-y plane, starting at the origin with initial velocity $5.00\ \mathrm{ms}^{-1}$ at an angle $40.0°$ above the positive x-axis. It was subjected to a constant acceleration of $3.00\ \mathrm{ms}^{-2}$ at an angle $80.0°$ above the negative x-axis. After 5.00 s, how far away was the object from the origin?

   A. 54.5 m
   B. 37.5 m
   C. 53.0 m
   D. 12.6 m
   E. 62.5 m
   F. 23.8 m
   G. 67.3 m
   H. 44.8 m

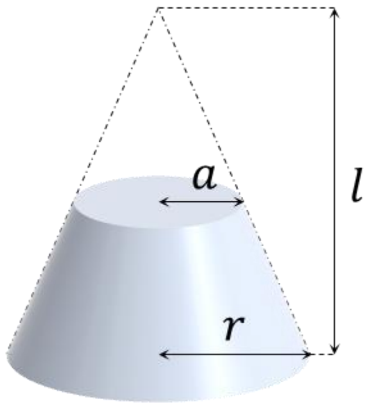
6. A man is in one of the capsules of a Ferris wheel. He conducts a drop test by releasing an object from his hand and allowing it to fall to the floor of the capsule. At which point during the revolution of the Ferris wheel would the object hit the floor in the least amount of time? You may assume the gravitational acceleration g does not change significantly with the Ferris wheel's height. You may also assume that the falling time is much shorter than the period of revolution of the Ferris wheel.

   A. When the capsule is moving vertically upwards.
   B. When the capsule is at the lowest point.
   C. When the capsule is moving vertically downwards.
   D. When the capsule is at the highest point.
   E. When the capsule is 45° from the bottom, on the way down
   F. When the capsule is 45° from the top, on the way down
   G. When the capsule is 45° from the bottom, on the way up
   H. The time taken to fall is the same at any point.

7. A particle of mass 10 g and charge $-500\ \mu\mathrm{C}$ initially moves horizontally with velocity $100\ \mathrm{ms}^{-1}$. It is subjected to a uniform electric field of $150\ \mathrm{NC}^{-1}$ vertically downwards. Ignoring gravity, what horizontal distance does the particle travel before it moves at an angle of 45° above the horizontal?

   A. 430 m
   B. 7500 m
   C. 750 m
   D. 6700 m
   E. 670 m
   F. 1300 m
   G. 130 m
   H. The particle's trajectory can never achieve this angle in finite time.

8. A circular frustum is made of a material with resistivity ρ. Its dimensions are as shown. The resistance between the two flat surfaces is R. Which of the following is a possible expression for R?



A. $\frac{\rho l}{\pi r^2}\left(\frac{1}{a}-\frac{1}{l}\right)$

B. $\frac{\rho l^2}{2\pi r}\left(\frac{1}{a}-\frac{1}{l}\right)$

C. $\frac{\rho l^2}{\pi r^2}\left(\frac{1}{a^2}-\frac{1}{l^2}\right)$

D. $\frac{\rho l}{\pi r^2}\left(\frac{1}{a^2}-\frac{1}{l^2}\right)$

E. $\frac{\rho l^2}{\pi r^2}\left(\frac{1}{a}-\frac{1}{l}\right)$

F. $\frac{\rho l^2}{\pi r}\left(\frac{1}{a^2}-\frac{1}{l^2}\right)$

G. $\frac{\rho l^2}{2\pi r^2}\left(\frac{1}{a^2}-\frac{1}{l^2}\right)$

H. $\frac{\rho l}{2\pi r^2}\left(\frac{1}{a^2}-\frac{1}{l^2}\right)$

9. Two long straight wires lie parallel to each other. They carry currents I_1 and I_2, and the current density is in the same direction. The distance between the two wires is d. What is the magnitude and direction of the magnetic force per unit length exerted on each wire?

   A. $\frac{\mu_0 I_1 I_2}{\pi d}$ Attractive
   B. $\frac{\mu_0 I_1 I_2}{\pi d}$ Repulsive
   C. $\frac{\mu_0 I_1 I_2}{2\pi d}$ Attractive
   D. $\frac{\mu_0 I_1 I_2}{2\pi d}$ Repulsive
   E. $\frac{\mu_0 I_1 I_2}{2\pi d^2}$ Attractive
   F. $\frac{\mu_0 I_1 I_2}{\pi d^2}$ Attractive
   G. $\frac{\mu_0 I_1 I_2}{\pi d^2}$ Repulsive
   H. None of the above

10. An object falling vertically from a great height is damped by air resistance and reaches terminal velocity. After it hits the ground, it bounces back up elastically. Which of the following statements is **true**?

   A. The acceleration of the object as it hits the ground is equal to the gravitational acceleration, g.
   B. If the mass of the object is doubled but its shape kept the same, the speed of the object as it bounces off the ground will remain the same.
   C. If the height from which the object is dropped is increased, the speed of the object as it bounces off the ground will increase.
   D. The acceleration of the object as it bounces off the ground is equal to twice the gravitational acceleration, $2g$.
   E. The acceleration of the object as it bounces off the ground is equal to the gravitational acceleration, g.
   F. The speed at which it bounces off the ground will be less than the terminal velocity.
   G. The speed at which it hits the ground will be less than the terminal velocity.
   H. None of the above

11. 2 hollow concentric spheres made of conducting material are set up such that the inner sphere has potential V_1 and the outer sphere has potential V_2, where $V_1 < V_2$. Which of the following statements is **false**?

   A. The electric field in between both spheres always points inwards.
   B. The 2 spheres form a capacitor.
   C. The electric potential inside the inner sphere is 0.
   D. The electric potential energy in this system is proportional to $(V_2 - V_1)^2$.
   E. A & B
   F. B & C
   G. C & D
   H. None of the above

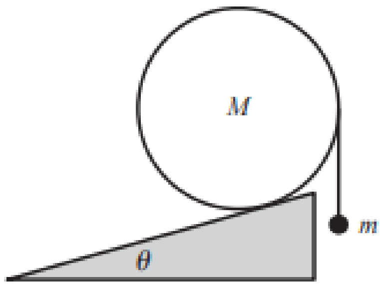
12. A solid sphere with mass M and radius R is given an impulse of J through its centre. Find the final speed of the sphere when it rolls without slipping.

A. J/M
B. $2J/5M$
C. $2J/3M$
D. $3J/5M$
E. $J/7M$
F. $J/5M$
G. $3J/7M$
H. $5J/7M$

13. A wheel is rotating with an initial angular velocity of $\omega_0 = 5\ \mathrm{rad}/s$. It is experiencing a constant angular acceleration of $\alpha = 3\ \mathrm{rad/s^2}$. After the wheel has rotated by 4 rad, what is the angular velocity of the wheel?

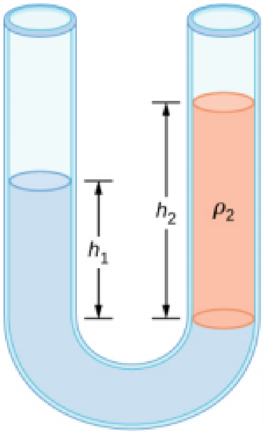
A. $7\ \mathrm{rad/s}$
B. $5\ \mathrm{rad/s}$
C. $1\ \mathrm{rad/s}$
D. $12\ \mathrm{rad/s}$
E. $17\ \mathrm{rad/s}$
F. $3\ \mathrm{rad/s}$
G. $9\ \mathrm{rad/s}$
H. $11\mathrm{rad/s}$

14. A ball of mass M rests on a rough incline of angle θ. A string is tied to the rightmost point of the ball with a mass hanging from it. Let the magnitude of normal force and friction force between slope and ball be N and f respectively. Let the magnitude of tension force and gravitational force of the ball be T and G respectively. Which of the following is a valid equation for torque balance?



A. $fR\cos\theta - TR\sin\theta + GR\sin\theta = 0$
B. $GR + fR\sin\theta - NR\cos\theta = 0$
C. $NR\cos\theta + fR - TR = 0$
D. $TR\cos\theta - fR\sin\theta = 0$
E. $TR\sin\theta - fR\cos\theta = 0$
F. $TR - fR = 0$
G. $TR - fR - GR = 0$
H. $NR + TR - fR - GR = 0$

15. Consider the U tube shown below. It is filled with 2 liquids of densities ρ_1 and ρ_2. What is the relationship between h_1 and h_2?



A. $\rho_2^2 h_1 = \rho_1^2 h_2$

B. $\left(1 + \frac{\rho_1}{\rho_2}\right) h_1 = \left(1 + \frac{\rho_2}{\rho_1}\right) h_2$

C. $\rho_1^2 h_1 = \rho_2^2 h_2$

D. $\rho_2 h_1^2 = \rho_1 h_2^2$

E. $\left(1 + \frac{\rho_2}{\rho_1}\right) h_1 = \left(1 + \frac{\rho_1}{\rho_2}\right) h_2$

F. $\rho_1 h_1 = \rho_2 h_2$

G. $\rho_2 h_1 = \rho_1 h_2$

H. $h_1 = h_2$

16. An airtight vessel has a volume ten times that of a container. Suppose that the pressure of the vessel is p_0. We perform the following process ten times: First connect the container to a source of constant pressure p_s. Then disconnect the container and connect it to the vessel. After the pressures have equalised, disconnect the container from the vessel. After finishing the processes, the pressure in the vessel is now $10p_0$. Find the ratio p_s/p_0. Assume that both the vessel and the container are kept at the same temperature as the surroundings.

A. 18.5
B. 14.9
C. 14.0
D. 16.6
E. 17.3
F. 16.2
G. 15.6
H. 17.6

17. Smoke emits from a chimney at a temperature of 40.0 °C. Assume that the smoke is an ideal gas of molar mass $\mu = 29.0$ g/mol and has molar specific heat at constant volume $c_v = 2.50R$. Given that the atmospheric pressure is uniform at 1 atm and the air temperature is uniform at 20.0 °C, calculate the height the smoke column rises. Neglect any heat exchange between the smoke and the surrounding air. **[Standard atmospheric pressure, $P_0 = 1.01 \times 10^5$ Pa]**

   A. 1780 m
   B. 550 m
   C. 726 m
   D. 1520 m
   E. 311 m
   F. 982 m
   G. 250 m
   H. 2150 m

18. The mass of Planet X is $M = 6.39 \times 10^{23}$ kg, and it has radius $r = 3.39 \times 10^6$ m. It is surrounded by an atmosphere of constant density, which consists of a gas of molar mass $\mu = 44.0$ g/mol. The height of the atmosphere is $h = 1.08 \times 10^4$ m. Determine the temperature of the atmosphere on the surface of Planet X.
   [Universal gravitational constant, G = 6.67×10^{-11} kg^{-1} m^3 s^{-2}]

   A. 61.0 K
   B. 120 K
   C. 189 K
   D. 131 K
   E. 267 K
   F. 243 K
   G. 198 K
   H. 212 K

19. Consider a circular beaker filled with liquid. The beaker is now spun at a constant angular velocity ω about its centre. What is the shape of the surface of the liquid?

   A. It will remain flat.
   B. It will be a cone.
   C. It will be a hyperbola.
   D. It depends on the magnitude of ω.
   E. It depends on the surface tension of the liquid.
   F. It will be a parabola.
   G. It depends on the viscosity of the liquid.
   H. It will be hemispherical.

20. Which of the following statements is true about capillary action?

   A. The height of the liquid column depends on the viscosity of the liquid.
   B. Increasing the surface tension will always increase the height of the liquid column.
   C. The height of the liquid column can be negative.
   D. Capillary action works even when there is no gravity.
   E. The radius of the tube has no effect on the height of the liquid column.
   F. A & B
   G. D & E
   H. B & D

21. Johnny is observing a rotating wheel experiencing a constant angular acceleration and measuring its angular speed. The table below shows the results he obtained.

Time (s)	Angular speed (rad s^{-1})
0	0.34
2	0.72
7	3.37

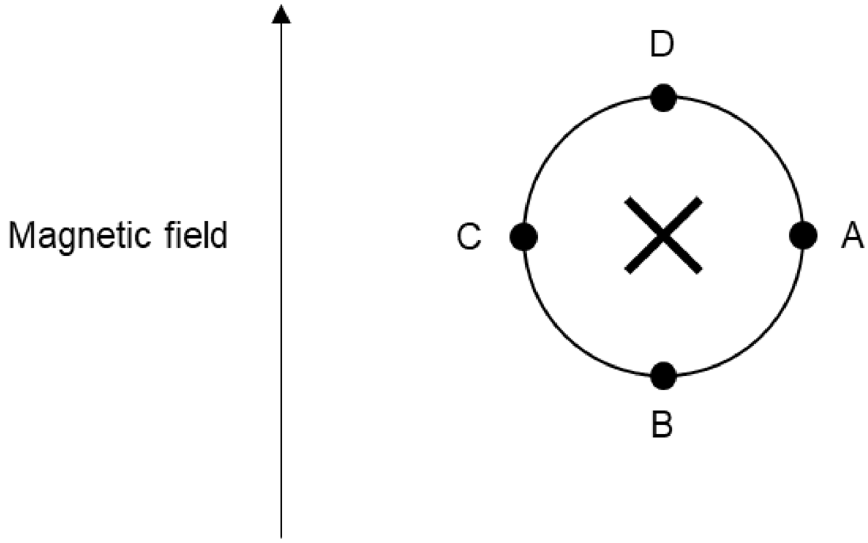
Find the value of the angular acceleration.

A. 0.19 rad s^{-2}
B. 0.36 rad s^{-2}
C. 0.43 rad s^{-2}
D. 0.65 rad s^{-2}
E. 0.34 rad s^{-2}
F. 0.53 rad s^{-2}
G. 0.48 rad s^{-2}
H. This scenario is impossible.

22. A non-uniform rod has length L and total mass M. The moment of inertia of the rod about an axis perpendicular to the rod through its geometrical centre and through the right end of the rod is $0.14ML^2$ and $0.19ML^2$ respectively. Find the position of its centre of mass from the left of the rod.

A. $0.72L$
B. $0.40L$
C. $0.64L$
D. $0.60L$
E. $0.67L$
F. $0.33L$
G. $0.45L$
H. $0.70L$

23. A stream of positively charged ions through a long straight circular tube which lines in a uniform magnetic field. The directions of the current and magnetic field are as shown. At which point is the electric potential lowest?



A. A
B. B
C. C
D. D
E. A & C
F. B & D
G. The centre of the circle.
H. The potentials at all points are equal.

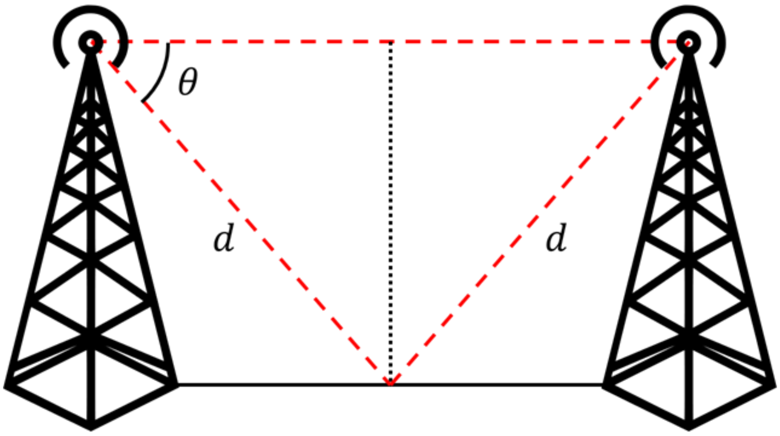
24. Some light is incident on a boundary between media of different refractive indices, at an angle $30°$ from the normal. The light then emerges at an angle $45°$ from the normal. What is the ratio of the speeds of the incident light to the emerging light?

A. $1: 2$
B. $2: 1$
C. $\sqrt{2}: 1$
D. $1: \sqrt{2}$
E. $1: 2\sqrt{2}$
F. $2\sqrt{2}: 1$
G. $1: 1$
H. None of the above

25. A battery is connected in series with a capacitor and an inductor, causing the current to oscillate with period T. The system is then scaled up by a factor of 2 in all directions. The number of coils in the inductor is kept constant. What is the new period of oscillation?

A. T
B. $\sqrt{2}\,T$
C. $2\sqrt{2}\,T$
D. $2\,T$
E. $4\,T$
F. $8\,T$
G. $0.5\,T$
H. None of the above.

26. Two antennas are set up as shown in the diagram below. A radio wave of wavelength $\lambda = 80.0$ m is sent from the left antenna to the right antenna. The wave takes the 2 paths as shown in the diagram in red. If $d = 10.0$ m, which of the following values of θ will result in the maximum signal strength being detected at the right antenna?



A. 30.0°
B. 36.9°
C. 23.0°
D. 53.1°
E. 45.0°
F. 36.0°
G. 60.0°
H. 55.0°

27. A capacitor is made of two $a \times a$ square plates, separated vertically by a distance a. Initially, a dielectric slab of dielectric constant κ and dimensions $a \times a \times 0.1a$ is slotted into the capacitor vertically. The capacitor is then connected to a voltage source, causing it to accumulate a charge Q_1. While still connected to the voltage source, the slab is then flipped such that it is now lying horizontally between the plates. The new charge across the capacitor is Q_2. Which of the following values of κ gives the smallest ratio Q_2/Q_1? Ignore any edge effects.

A. 5.00
B. 10.0
C. 20.0
D. 30.0
E. 50.0
F. 80.0
G. 100
H. 200

28. An electric dipole can be considered simply as two point charges $+Q$ and $-Q$ separated by a distance d. Let r denote the distance from the centre of the dipole. At very large distances ($r \gg d$) along the direction of the axis of the dipole, what is the approximate relation between the magnitude of the electric field E and r? [Hint: If $x \ll 1$, then $(1+x)^n \approx 1 + nx$.]

A. $E \propto r$

B. $E \propto \frac{1}{r}$

C. $E \propto \frac{1}{r^2}$

D. $E \propto \frac{1}{r^4}$

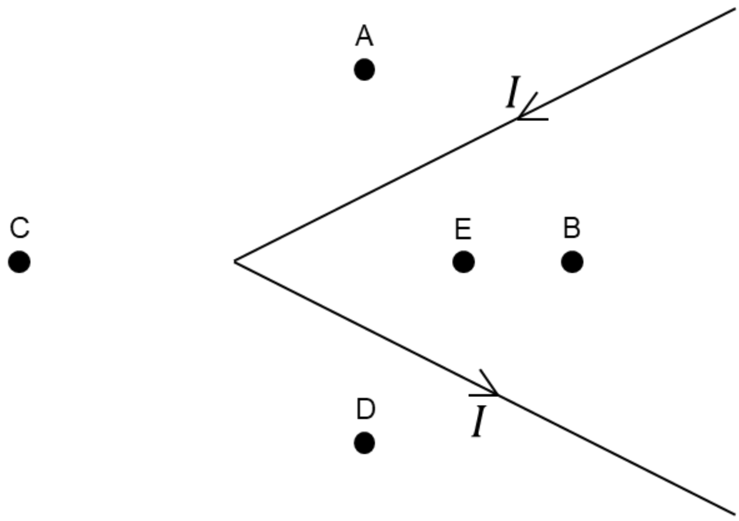
E. $E \propto r^2$

F. $E \propto \frac{1}{r^3}$

G. $E \propto r^3$

H. E is independent of r.

29. A kinked semi-infinite wire (i.e., the wires extend infinitely to the right) lies in the plane and contains a current as shown. E is obtained by reflecting A about the top wire. D is obtained by reflecting E about the bottom wire. C is obtained by reflecting E about the kink. Points C, E and B are collinear, and B is further from the kink than E. At which point is the magnitude of the magnetic flux density the greatest?



A. A
B. B
C. C
D. D
E. E
F. A& D
G. B & E
H. A & D & C

30. On a level surface, a car is able to accelerate at up to $0.30g$ and brake at up to $0.70g$. Consider a stretch of road inclined at 10°, with length 200 m. Find the minimum time required for a car to start from rest, accelerate up the incline and come to a stop at the 200 m mark.
[Acceleration due to gravity at the surface of Earth, $\boldsymbol{g = 9.81\ \mathrm{ms^{-2}} = |g|}$]

A. 10 s
B. 19 s
C. 14 s
D. 17 s
E. 13 s
F. 22 s
G. 27 s
H. 24 s

31. You have seven point charges of charge $+Q$ and two point charges of charge $-Q$. You wish to arrange these nine point charges to form a regular nine-sided polygon of side length L. What is the largest possible magnitude of electric field at the centre of the polygon?

A. $0.25\frac{Q}{\pi\varepsilon_0 L^2}$

B. $0.50\frac{Q}{\pi\varepsilon_0 L^2}$

C. $0.47\frac{Q}{\pi\varepsilon_0 L^2}$

D. $3.8\frac{Q}{\pi\varepsilon_0 L^2}$

E. $0.76\frac{Q}{\pi\varepsilon_0 L^2}$

F. $2.5\frac{Q}{\pi\varepsilon_0 L^2}$

G. $4.4\frac{Q}{\pi\varepsilon_0 L^2}$

H. $0.44\frac{Q}{\pi\varepsilon_0 L^2}$

32. A ring of radius R carries uniformly distributed charge $+Q$. A point charge $-q$ is located at distance H from the centre of the ring, perpendicular to the plane of the ring. What is the force exerted by the charge on the ring?

A. $\frac{1}{4\pi\varepsilon_0}\frac{QqH}{(H^2+R^2)^{3/2}}$ towards the charge

B. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H^2+R^2}$ away from the charge

C. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H^2+R^2}$ towards the charge

D. $\frac{1}{4\pi\varepsilon_0}\frac{QqH}{(H^2+R^2)^{3/2}}$ away from the charge

E. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H^2}$ towards the charge

F. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H^2}$ away from the charge

G. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H+R}$ away from the charge

H. $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{H+R}$ towards the charge

33. A thin wire made of an insulating material is bent into the shape of an equilateral triangle with side length L, and placed in a region with a uniform electric field E. What is the maximum possible electric potential difference between any 2 points on the wire?

A. $\frac{\sqrt{3}}{2}EL$

B. $\frac{2}{3}EL$

C. $\sqrt{3}EL$

D. EL

E. $\frac{\sqrt{3}}{3}EL$

F. $2EL$

G. $3EL$

H. None of the above

34. A mass m is suspended vertically from the ceiling by a uniform elastic rod of spring constant k. The system is initially at equilibrium, and the rod has length L. The system is then instantly scaled up by a factor of 2 in all directions (length, breadth, and height), and it starts to oscillate. Given that the spring constant of a uniform rod is directly proportional to its cross-sectional area, find the maximum speed achieved by this oscillating system.

A. $g\sqrt{\frac{m}{k}}$

B. $4g\sqrt{\frac{m}{k}}$

C. $2g\sqrt{\frac{m}{k}}$

D. $g\sqrt{\frac{2m}{k}}$

E. $2g\sqrt{\frac{2m}{k}}$

F. $4g\sqrt{\frac{2m}{k}}$

G. $2g\sqrt{\frac{m}{2k}}$

H. None of the above

35. 3 polarizers, A, B and C, are placed in that order, parallel to each other. Some initially unpolarized light of intensity I_0 is shone perpendicularly through A. Given that polarizer C is angled at 90° with respect to A, what is the maximum intensity of light emerging from C that can be detected?

A. I_0

B. $\frac{1}{2}I_0$

C. $\frac{1}{8}I_0$

D. $\frac{1}{4}I_0$

E. $2I_0$

F. $\frac{1}{3}I_0$

G. $\frac{2}{3}I_0$

H. None of the above

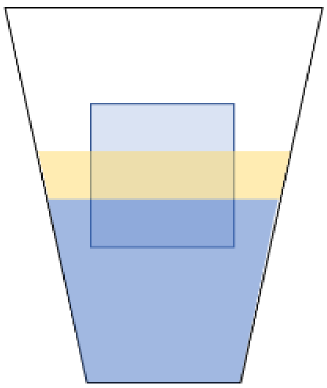
36. A circular disc of radius $R = 1.00$ cm and mass $m = 50.0$ g is resting on the surface of a liquid, held up by surface tension. What is the minimum value of surface tension for the disc to be supported? **[Acceleration due to gravity at the surface of Earth, $\boldsymbol{g = 9.81\ \mathrm{ms}^{-2} = |g|}$]**

A. $1560\ \mathrm{Nm}^{-1}$
B. $24.5\ \mathrm{Nm}^{-1}$
C. $49.1\ \mathrm{Nm}^{-1}$
D. $156\ \mathrm{Nm}^{-1}$
E. $7.81\ \mathrm{Nm}^{-1}$
F. $78.1\ \mathrm{Nm}^{-1}$
G. $1.56\ \mathrm{Nm}^{-1}$
H. $2.45\ \mathrm{Nm}^{-1}$

37. Consider a wind turbine with blades of length $l = 80$ m. Suppose that the wind speed in the area is $v = 8.0$ m/s and the air has density $\rho = 1.3\ \mathrm{kg/m^3}$. The efficiency of the turbine is $\eta = 37$ %. What is the power output of the turbine?

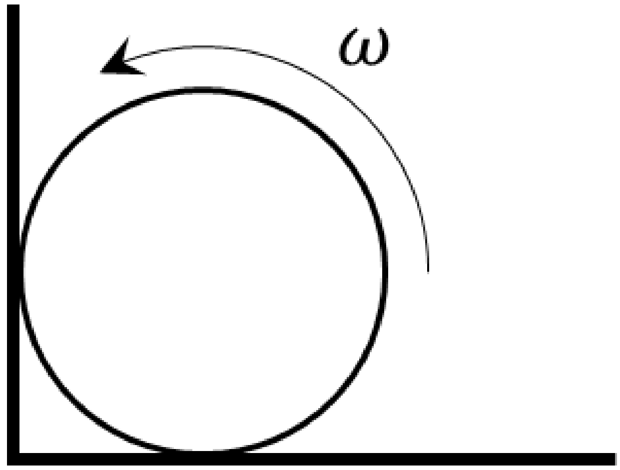
A. 2.5 MW
B. 300 kW
C. 500 kW
D. 30 kW
E. 6.7 MW
F. 5.0 MW
G. 670 kW
H. 7.5 MW

38. A block of density ρ is floating in a cup of water and oil, which has densities $\rho_w = 1000\ \mathrm{kgm}^{-3}$ and $\rho_o = 750\ \mathrm{kgm}^{-3}$ respectively. If a third of the cube is in the water and a third of it is in oil, what is ρ?



A. $500\ \mathrm{kgm}^{-3}$
B. $750\ \mathrm{kgm}^{-3}$
C. $630\ \mathrm{kgm}^{-3}$
D. $580\ \mathrm{kgm}^{-3}$
E. $830\ \mathrm{kgm}^{-3}$
F. $880\ \mathrm{kgm}^{-3}$
G. $250\ \mathrm{kgm}^{-3}$
H. $480\ \mathrm{kgm}^{-3}$

39. A solid cylinder with mass M and radius R is rotating against a rough wall. It has initial angular speed ω and the coefficient of friction between the cylinder and wall is μ. Find the time taken for the cylinder to stop rotating.



A. $\frac{R\omega}{2g\mu}$

B. $\frac{R\omega\mu}{2g(1+\mu^2)}$

C. $\frac{R\omega(1+\mu^2)}{g\mu(1+\mu)}$

D. $\frac{R\omega}{2g(1+\mu)}$

E. $\frac{R\omega(1+\mu)}{2g\mu(1+\mu^2)}$

F. $\frac{R\omega(1+\mu^2)}{2g\mu}$

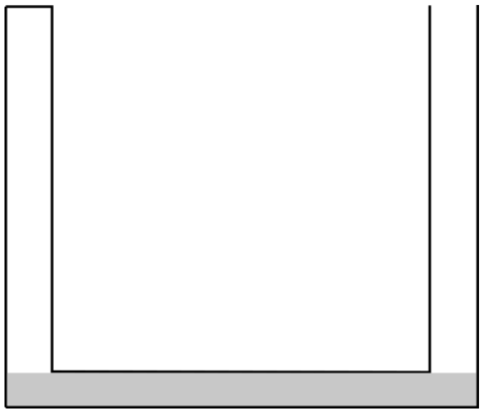
G. $\frac{R\omega(1+\mu^2)}{2g\mu(1+\mu)}$

H. $\frac{R\omega}{g\mu(1+\mu)}$

40. A container contains a mixture of hydrogen and helium of total mass 140 g and is closed by a light piston that is free to move with negligible friction. A quantity of heat $Q = 110$ kJ is transferred to the gas at constant pressure. The gas performs a work of 40 kJ. Calculate the mass of hydrogen in the mixture.

A. 43 g
B. 15 g
C. 61 g
D. 70 g
E. 120 g
F. 87 g
G. 30 g
H. 20 g

41. A U-shaped tube is sealed at one end and consists of three arms each of length $l = 0.30$ m and cross-sectional area $A = 5.0$ cm^2. The air in the sealed arm is heated and displaces the mercury out of the U-shaped tube. Find the work done by the gas. The atmospheric pressure is 1.013×10^5 Pa = 760 mmHg.
[Acceleration due to gravity at the surface of Earth, $g = 9.81$ ms^{-2} = $|g|$]



A. 6.0 J
B. 20 J
C. 3.6 J
D. 87 J
E. 120 J
F. 12 J
G. 36 J
H. 72 J

42. 2 equally charged particles of 1.00 mC and mass 1.00 kg are moving towards each other from afar at a speed of 30.0 ms^{-1} and 50.0 ms^{-1} respectively. Find the minimum distance of approach.
[Permittivity of free space, $\varepsilon_0 = 8.85 \times 10^{-12}$ C^{2}N^{-1}m^{-2}]

A. 0.096 m
B. 0.089 m
C. 0.189 m
D. 0.178 m
E. 0.378 m
F. 0.756 m
G. 0.269 m
H. 0.576 m

43. Consider an infinite number of charges along the x-axis. $q_0 = 1.00$ nC is at $x = 1.00$ m, $q_1 = 2.00$ nC is at $x = 2.00$ m, $q_2 = 4.00$ nC is at $x = 4.00$ m and so on. More generally, $q_i = 2.00^i$ nC at $x = 2.00^i$ m. Find the electric field at $x = 0.00$ m.
[Permittivity of free space, $\varepsilon_0 = 8.85 \times 10^{-12}$ C^{2}N^{-1}m^{-2}]

A. -8.99 NC^{-1}
B. 8.99 NC^{-1}
C. 4.49 NC^{-1}
D. -4.49 NC^{-1}
E. 0.00 NC^{-1}
F. 2.25 NC^{-1}
G. -2.25 NC^{-1}
H. 1.00 NC^{-1}

44. A mass m is suspended from the ceiling by an inelastic, uniform string of mass m and length L. A wave pulse of length $\Delta x \ll L$ is sent upwards along the string from the position of the mass. What is the new length of the pulse $\Delta x'$ when it hits the ceiling?

A. $2\frac{(\Delta x)^2}{L}$

B. $2\Delta x$

C. $\sqrt{2}\frac{(\Delta x)^2}{L}$

D. $\sqrt{2}\Delta x$

E. $2\sqrt{2}\frac{(\Delta x)^2}{L}$

F. $2\sqrt{2}\Delta x$

G. $\frac{\Delta x}{\sqrt{2}}$

H. None of the above

45. An insulated container of length $2\ell = 1.0$ m has a spring of natural length $\frac{\ell}{2}$ attached to one end, and a frictionless piston attached to the other end of the spring dividing the container into two parts. Each of the parts has 1 mole of helium at temperature $T = 300$ K. The spring is initially at an extension of $x_0 = 0.12$ m. A hole is then made in the piston and the gas can flow between the two compartments. Find the change in temperature of the gas.

A. 2.6 K
B. 9.6 K
C. 5.3 K
D. 10 K
E. 13 K
F. 3.3 K
G. 4.2 K
H. 6.7 K

46. 10 double slits, all with identical distance between slits w, are placed back-to-back, parallel to each other, each a distance $d = 1.0$ m apart. Light of wavelength 500 nm is then shone normally onto the first double slit in the sequence. A detector is placed a distance d behind the center of the 10th double slit. Which of the following values for w will maximize the intensity of light detected?

A. 1.2 mm
B. 1.8 mm
C. 1.6 mm
D. 2.0 mm
E. 1.4 mm
F. 1.0 mm
G. 2.2 mm
H. 1.5 mm

47. A uniformly charged solid insulating cube has electric potential V_0 at its center and V_1 at its corners. What is the ratio V_0/V_1?

A. 4
B. 2
C. 8
D. 16
E. 10
F. 12
G. 32
H. None of the above

48. A spherical ball of some radius is placed into a vertical hollow tube of length L and identical radius that is closed at the other end. The ball slowly sinks into the tube, until its center is $L/2$ above the bottom of the tube, where it comes to rest. The ball is then gently pushed, causing it to oscillate about the halfway point on the tube. Approximately what is the period of the ball's oscillation?

A. $2\pi\sqrt{\frac{L}{g}}$

B. $\pi\sqrt{\frac{2L}{g}}$

C. $\pi\sqrt{\frac{L}{2g}}$

D. $\pi\sqrt{\frac{4L}{g}}$

E. $\pi\sqrt{\frac{L}{4g}}$

F. $\pi\sqrt{\frac{L}{g}}$

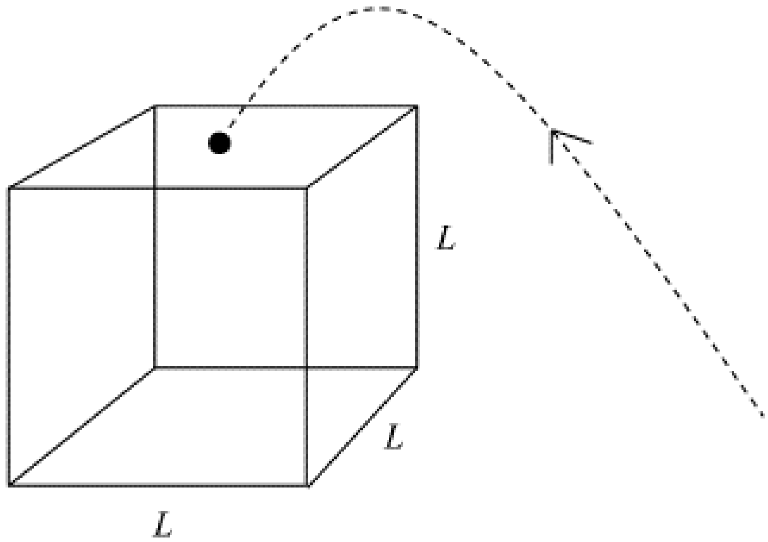
G. $\frac{\pi}{2}\sqrt{\frac{L}{g}}$

H. None of the above

49. A ball of mass m is in between 2 identical objects of mass M. You then give the ball an initial velocity v towards one of the mass M. The ball collides elastically with that mass M before bouncing back and colliding with the other mass and then bouncing back. What is the maximum mass of m in terms of M such that it will collide with the first mass M a second time?

A. M
B. $4.23M$
C. $0.236M$
D. $0.472M$
E. $2.12M$
F. $2.36M$
G. $0.118M$
H. $0.667M$

50. Consider a large cube-shaped building with side length L. From the ground, you wish to launch a projectile such that it lands at the centre of the top of the building, without hitting or bouncing off other parts of the building prior to landing. Neglecting any effects of drag, what is the minimum launch speed required to do so?



A. $\sqrt{gL}$

B. $\sqrt{2gL}$

C. $\sqrt{3gL}$

D. $\sqrt{5gL}$

E. $\sqrt{\frac{5}{2}gL}$

F. $\sqrt{\frac{3}{2}gL}$

G. $\sqrt{\frac{7}{2}gL}$

H. $\sqrt{7gL}$

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